
Weather-Conditioned Deep Learning for Bird Migration Trajectory Forecasting

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Abstract

Forecasting animal migration from GPS traces is difficult because most daily observations are stationary or short stopover movements, while a small fraction are long directed flights. We study next-day southbound bird trajectory forecasting from recent movement history and optional weather covariates. Our final experiment compares direct displacement regression against a triline decomposition into fly/no-fly, distance, and heading; compares MLP, Transformer, and LSTM encoders; and evaluates context lengths $k \in \{7, 14, 30\}$ under a year-grouped 80/20 train/test split. Across the completed experiment, a 4-layer triline LSTM is the strongest overall model, reaching 17.03 km mean error and 51.57 km P90 error at $k = 30$ with weather. A larger Transformer with a CLS token is competitive at long context but does not consistently dominate the LSTM. Weather covariates provide only small and inconsistent gains under year-held-out evaluation, and autoregressive rollout experiments show that one-step errors compound substantially over full migration paths.

1 Introduction

Bird migration prediction is a spatiotemporal forecasting problem with direct ecological and engineering relevance. Given a short window of daily GPS observations for an individual migration path, the goal is to predict the bird’s next-day latitude and longitude. Accurate forecasts can support migration ecology, risk assessment for human infrastructure, and analysis of how environmental conditions shape large-scale movement. The task is challenging because migration paths alternate between stopovers and long flights: a model must decide whether the bird will move, how far it will move, and in which direction.

This project studies southbound migration trajectories with aligned weather observations. We make three empirical contributions. First, we compare direct coordinate regression with a triline movement decomposition that predicts fly/no-fly, distance, and heading before reconstructing coordinates. Second, we evaluate whether a Transformer encoder improves over a recurrent LSTM and a feed-forward MLP baseline for short-to-medium context windows. Third, we test whether weather covariates improve generalization when train and test examples come from different years.

Our main finding is that the simplest “newer is better” story does not hold. The triline LSTM is the most reliable model across metrics and converges faster than the larger Transformer. The Transformer V2 becomes competitive at $k = 30$, suggesting that attention is useful only when enough temporal context is available. Weather features do not reliably improve held-out-year accuracy, indicating that weather joins and interannual distribution shift remain important limitations.

2 Related Work

Animal movement forecasting builds on both ecological modeling and sequence learning. Large tracking repositories such as Movebank organize animal GPS records, individual metadata, and study structure, making multi-year trajectory analysis feasible at scale Kranstauber et al. (2011). Classical movement ecology often models paths probabilistically rather than as supervised prediction; dynamic Brownian bridge movement models, for example, estimate utilization distributions from observed tracks and time-varying movement variance Kranstauber et al. (2012). These methods are interpretable and biologically grounded, but they are not designed to learn nonlinear effects from high-dimensional temporal covariates.

Deep sequence models offer a complementary approach. Recurrent networks such as LSTMs encode ordered history with a strong recency bias, which is attractive for next-step movement prediction. Transformers replace recurrence with self-attention and have been widely adapted to long-horizon time-series forecasting. Informer introduced efficient sparse attention for long sequence forecasting Zhou et al. (2021), while Autoformer combined decomposition with attention-like autocorrelation mechanisms Wu et al. (2021). These models motivate our Transformer baseline, but our setting differs in two ways: the data are smaller and noisier than common benchmark time series, and the output distribution is highly imbalanced between stationary days and migration flights. Our triline formulation explicitly targets that imbalance by separating movement occurrence, magnitude, and direction.

3 Data

We use a cleaned southbound bird migration dataset with matched weather features. Each row is a daily path observation. The final dataset contains 13,550 daily records, 135 migration paths, and 71 source birds spanning 2013-07-30 to 2023-12-26. For each context length k , a training example consists of the previous k daily observations, and the target is the next-day GPS location.

Table 1: Dataset and split summary.

Field	Value
Daily records	13,550
Migration paths	135
Source birds	71
Date range	2013-07-30 to 2023-12-26
Context lengths	$k = 7, 14, 30$ days
Train/test protocol	Year-grouped 80/20 split by target year
Prediction target	Next-day latitude and longitude
Fly threshold	10 km daily displacement

The no-weather setting uses location, displacement, step length, heading, day-of-year encoding, and stopover duration. The weather setting augments these with temperature, wind speed and direction, surface pressure, log precipitation, cloud cover, and boundary layer height. All feature normalization statistics are fit only on the training windows for each k .

4 Methods

4.1 Problem Formulation

For a path with daily feature vectors $\mathbf{x}_{t-k+1:t}$ and current location $\mathbf{s}_t = (\text{lat}_t, \text{lon}_t)$, we predict the next location $\hat{\mathbf{s}}_{t+1}$. Evaluation uses haversine distance between $\hat{\mathbf{s}}_{t+1}$ and the observed $\mathbf{s}_{t+1}$. We report mean error, median error, P90 error, and migration-day error for days with at least 50 km displacement.

4.2 Direct Regression

The direct models predict normalized coordinate displacement:

$$\widehat{\Delta \mathbf{s}}_{t+1} = [\widehat{\Delta \text{lat}}_{t+1}, \widehat{\Delta \text{lon}}_{t+1}], \quad \widehat{\mathbf{s}}_{t+1} = \mathbf{s}_t + \widehat{\Delta \mathbf{s}}_{t+1}.$$

This target is simple and optimized directly for coordinate movement, but it does not expose whether the model understands the fly/stationary decision.

4.3 Triline Decomposition

The triline models predict three movement components from an encoded history representation $\mathbf{h}$:

$$\widehat{p}_{fly} = \sigma(f_{fly}(\mathbf{h})), \quad (1)$$

$$\widehat{d} = \exp(f_{dist}(\mathbf{h})) - 1, \quad (2)$$

$$[\widehat{\sin \theta}, \widehat{\cos \theta}] = f_{dir}(\mathbf{h}). \quad (3)$$

The model is trained with binary cross entropy for the 10 km fly label, mean squared error on log distance, and mean squared error on heading sine/cosine. At inference, distance and heading are converted into latitude/longitude displacement using a local spherical approximation. We also evaluate gated triline predictions, where predicted distance is set to zero when $\widehat{p}_{fly}$ is below the selected threshold.

4.4 Architectures

We compare four model families. The Direct MLP flattens the k -day input window and predicts displacement with feed-forward layers. The Direct Transformer 4L uses a Transformer encoder for the same displacement target. The Triline LSTM 4L uses a 4-layer LSTM and triline heads. The Triline Transformer V2 uses a CLS token, 4 Transformer encoder layers, $d_{model} = 256$, 8 attention heads, GELU feed-forward blocks, and triline heads. Bird identity embeddings were disabled in the final runs to avoid incompatibility with year-grouped splitting and to emphasize generalization from path dynamics rather than memorized individuals.

5 Experimental Setup

All final experiments use seed 42, CUDA, AdamW with learning rate 10^{-3} and weight decay 10^{-4} , batch size 128, maximum 200 epochs, and early stopping patience 30. The learning rate scheduler is ReduceLROnPlateau with factor 0.5 and patience 4. The year-grouped split assigns the earliest 80% of target years to training and the latest 20% to testing, preserving annual cycles while reducing temporal leakage.

Table 2: Model sizes in the final experiment. Parameter counts vary slightly with feature mode and context length; values shown are representative for $k = 30$.

Model	Parameters without weather	Parameters with weather
Direct MLP	47K	78K
Direct Transformer 4L	807K	808K
Triline LSTM 4L	2.16M	2.16M
Triline Transformer V2	3.30M	3.30M

6 Results

6.1 Next-Day Prediction

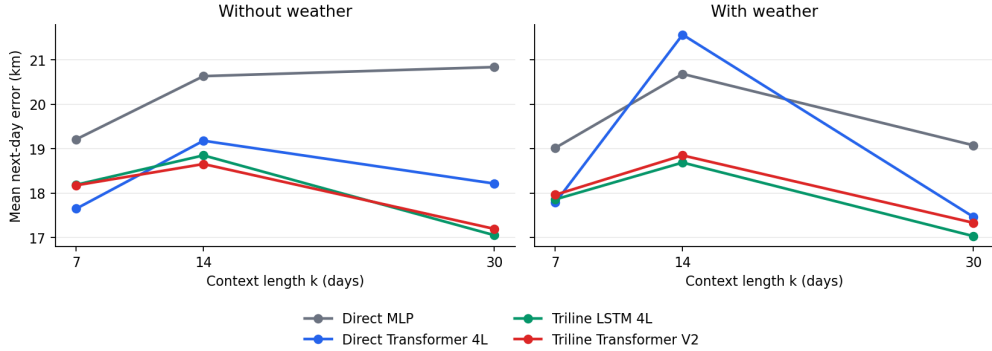


Figure 1: Mean next-day GPS error across context lengths. Longer context helps the triline models most clearly; the LSTM remains the strongest overall model, while Transformer V2 becomes competitive at $k = 30$.

Table 3: Next-day prediction results for $k = 30$. Lower error is better.

Condition	Model	Mean km	Median km	P90 km	Mig.50 mean km
No weather	Direct MLP	20.84	5.75	62.48	85.35
No weather	Direct Transformer 4L	18.22	4.35	57.93	81.35
No weather	Triline LSTM 4L	17.06	2.45	49.66	96.95
No weather	Triline Transformer V2	17.20	2.27	49.34	99.65
Weather	Direct MLP	19.08	5.02	58.84	93.30
Weather	Direct Transformer 4L	17.47	3.45	59.76	81.69
Weather	Triline LSTM 4L	17.03	2.47	51.57	96.24
Weather	Triline Transformer V2	17.33	2.42	52.78	99.56

At the longest context, triline models dominate median and P90 error, indicating better handling of typical stationary days and difficult tail cases. The Direct Transformer remains a strong lower-parameter baseline, especially on migration-day mean error in the weather setting. Overall, however, the triline LSTM has the best mean error with weather and is essentially tied for best mean without weather.

6.2 Architecture and Context Trends

Table 4: Best model by mean error for each context and feature setting.

Condition	k	Best mean-error model	Mean km
No weather	7	Direct Transformer 4L	17.65
No weather	14	Triline Transformer V2	18.66
No weather	30	Triline LSTM 4L	17.06
Weather	7	Direct Transformer 4L	17.79
Weather	14	Triline LSTM 4L	18.69
Weather	30	Triline LSTM 4L	17.03

The LSTM converges faster than Transformer V2: in the weather runs, LSTM best epochs are 4, 4, and 2 for $k = 7, 14, 30$, while Transformer V2 best epochs are 17, 30, and 10. This supports the

interpretation that the LSTM’s recency-biased inductive structure fits next-day migration forecasting well. Transformer V2 still improves relative to shorter contexts and is strongest in several median or tail-error slices at $k = 30$.

6.3 Weather Ablation

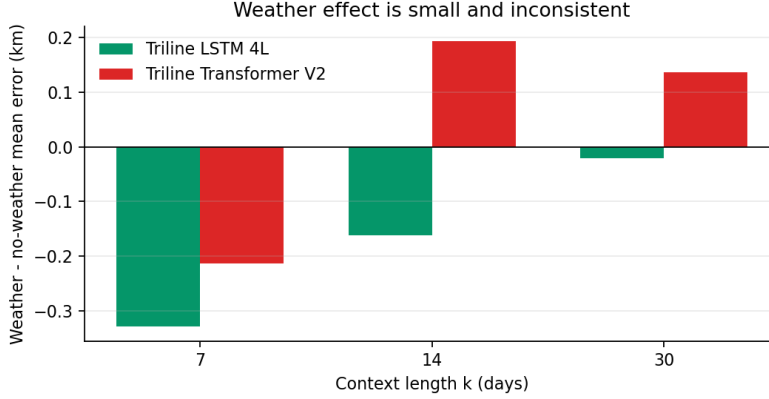


Figure 2: Weather minus no-weather mean error for triline models. Negative values mean weather helps. Weather provides small gains at $k = 7$ but inconsistent effects at longer context.

Table 5: Weather ablation for triline models. Delta is weather minus no-weather mean error; negative is better with weather.

Model	k	No-weather mean	Weather mean	Delta
Triline LSTM 4L	7	18.19	17.86	-0.33
Triline LSTM 4L	14	18.85	18.69	-0.16
Triline LSTM 4L	30	17.06	17.03	-0.02
Triline Transformer V2	7	18.18	17.96	-0.21
Triline Transformer V2	14	18.66	18.85	+0.19
Triline Transformer V2	30	17.20	17.33	+0.14

Weather does not provide the clear improvement we expected. It slightly improves both triline models at $k = 7$, but the effect nearly disappears for the LSTM and reverses for the Transformer at $k = 14$ and $k = 30$. A likely explanation is that year-held-out evaluation changes the weather distribution: models cannot simply reuse same-year seasonal/weather patterns, and the joined weather variables may not capture the departure-time conditions that trigger long flights.

6.4 Autoregressive Rollout

Next-day accuracy is necessary but not sufficient for full-path forecasting. We therefore evaluate autoregressive rollouts on 27 valid held-out paths, using at least 60 observed days before rolling forward. These rollouts repeatedly feed predictions back into the model context, so small one-step biases can grow into large path-level errors.

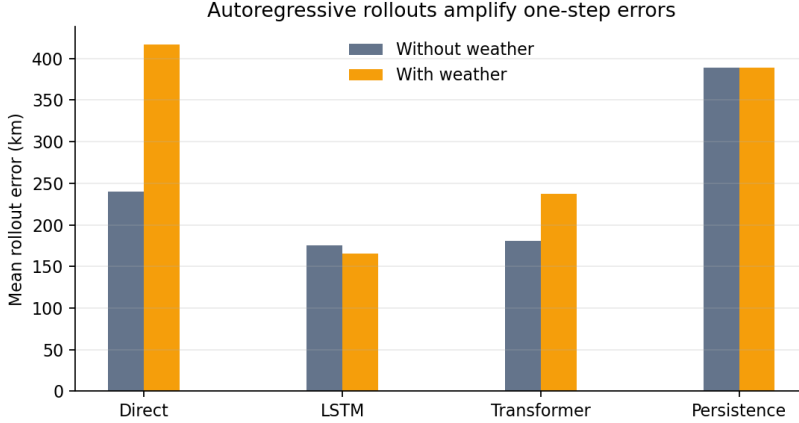


Figure 3: Mean autoregressive rollout error. Learned one-step models improve over persistence in some settings, but rollout remains much harder than one-day prediction.

Table 6: Autoregressive rollout aggregate errors over 27 valid paths.

Condition	Direct	LSTM	Transformer	Persistence
No weather mean km	240.07	175.34	180.84	388.62
Weather mean km	417.22	165.16	237.28	388.62
No weather final km	343.52	206.95	227.91	–
Weather final km	603.58	193.77	323.68	–

The LSTM is also the most reliable rollout model. Weather improves LSTM rollout error by about 10 km on average but worsens Direct and Transformer rollouts. Representative GIFs in `rollout/with_weather/gifs/` and `rollout/without_weather/gifs/` illustrate both successful paths and hard failures where early directional errors compound over the remainder of the migration.

7 Discussion

The direct-vs.-triline comparison shows a tradeoff. Direct regression is surprisingly strong and parameter-efficient, especially with a Transformer encoder. However, triline decomposition gives substantially lower median error and competitive or better tail error because it models the fly/stationary imbalance explicitly. Gating further reduces typical stationary-day error, though it can hurt large migration-day accuracy when the fly head misses a departure.

The architecture comparison favors the LSTM. The LSTM’s hidden state naturally emphasizes recent observations, which is valuable for next-day forecasting where the last few days often determine whether a bird is still in stopover or has resumed migration. Transformer V2 is more flexible and obtains strong median/P90 results at $k = 30$, but it is larger, slower to converge, and not consistently more accurate.

The weather ablation is the clearest limitation. Weather should matter biologically, especially wind and pressure, but the measured gains are weak under year-grouped testing. This suggests that our current weather features, daily aggregation, and split protocol make the weather signal hard to generalize. Future work should use departure-time weather, tailwind and crosswind relative to intended heading, stronger regularization for weather features, and explicit out-of-year validation.

8 Conclusion

We evaluated deep models for next-day southbound bird migration forecasting using GPS history and optional weather covariates. The best overall approach is a triline 4-layer LSTM, which

combines an appropriate sequential inductive bias with an interpretable movement decomposition. Transformer V2 is competitive for long contexts but does not justify its larger size across all metrics. Weather features provide only small and inconsistent benefits under a year-held-out split, and full-path rollouts reveal substantial compounding error. The most promising next steps are stronger weather alignment, uncertainty-aware rollouts, and hybrid models that combine triline interpretability with biologically informed movement constraints.

Reproducibility Checklist

Table 7: Artifacts used to reproduce the final report.

Item	Location or value
Dataset	<code>data/combined_southbound_paths_with_weather_matched.csv</code>
Final experiment script	<code>model_files/run_final_path_experiment.py</code>
Rollout/GIF script	<code>model_files/run_current_rollout_gifs.py</code>
Model definitions	<code>model_files/model.py</code>
Result summaries	<code>with_weather/</code> , <code>without_weather/</code> , <code>rollout/</code>
Random seed	42
Hardware target	CUDA GPU when available

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